

Rosen — §2.1: Sets

Selected Exercises

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Exercise 11

Determine whether each of these statements is true or false.

- (a) $x \in \{x\}$
- (b) $\{x\} \subseteq \{x\}$
- (c) $\{x\} \in \{x\}$
- (d) $\{x\} \in \{\{x\}\}$
- (e) $\emptyset \subseteq \{x\}$
- (f) $\emptyset \in \{x\}$

Exercise 17

Suppose that A , B , and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

Exercise 23

How many elements does each of these sets have where a and b are distinct elements?

- (a) $\mathcal{P}(\{a, b, \{a, b\}\})$
- (b) $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
- (c) $\mathcal{P}(\mathcal{P}(\emptyset))$

Exercise 31

Let A be a set. Show that $\emptyset \times A = A \times \emptyset = \emptyset$.

Exercise 37

How many different elements does A^n have when A has m elements and n is a positive integer?

Exercise 45

The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair (a, b) to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and

only if $a = c$ and $b = d$. [*Hint*: First show that $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$.]

Exercise 46

This exercise presents Russell's paradox. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.

- (a) Show the assumption that S is a member of S leads to a contradiction.
- (b) Show the assumption that S is not a member of S leads to a contradiction.

By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.